

Plan A + Physics Loss - Technical Report

Date: 2026-03-30

Model: PlanAPlusPhysics (Plan A + Physics Constraint Auxiliary Loss)

Status: Completed

1. Research Objective

Verify whether physics equations as auxiliary loss can improve the generalization performance of the Plan A large model on power extrapolation tasks.

Core Problem: Training data is 1-5 components (power 2.5-13.7W), testing is 6-9 components (power 15.7-30W). Pure data-driven models show severe performance degradation at boundaries.

2. Physics Equation Derivation

2.1 Heat Equation (Steady-State)

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Discretized as 5-point Laplacian:

$$\text{Laplacian} = T[i+1,j] + T[i-1,j] + T[i,j+1] + T[i,j-1] - 4 \cdot T[i,j] = 0$$

2.2 Boundary Condition (Robin BC)

$$-k \cdot \frac{dT}{dn} = h \cdot (T - T_{\text{amb}})$$

Discrete BC equation (edge node i=0):

$$(k/dx^2 + h/dx) \cdot T_{\text{edge}} - k/dx^2 \cdot T_{\text{adj}} - h/dx \cdot T_{\text{amb}} = 0$$

Parameters:

Parameter	Value	Description
-----	-----	-----
k_fr4	0.35 W/(mK)	FR4 substrate thermal con
h_conv	30 W/(m2K)	Convection heat transfer
dx_norm	1/99 = 0.0101	Normalized grid spacing
k_dx2	3430.35	k/dx^2
h_dx	2970.0	h/dx
a_edge	6400.35	k_dx2 + h_dx
c_adj	0.536	Normalized BC coefficient
c_amb	0.464	= h_dx/a_edge

2.3 Physics Normalization

`T_norm = (T - T_amb) / P_total`

Under physics_norm, BC simplifies to (no ambient term):

`T_norm_edge - c_adj * T_norm_adj = 0`

2.4 Heat Source Mask

`is_source[i,j] = 1 if min_dist(component_center, grid[i,j]) < 0.06 else 0`

PDE loss is only computed on interior non-source points.

3. Model Architecture

3.1 PlanAPlusPhysics Wrapper

```
class PlanAPlusPhysics(nn.Module):  
  
    def __init__(self, base_model, k_fr4=0.35, h_conv=30.0, dx_norm=1.0/99.0):  
  
        self.base = base_model # SetFN0Model  
  
        self.k_fr4 = k_fr4  
  
        self.h_conv = h_conv  
  
        k_dx2 = k_fr4 / (dx_norm ** 2)  
  
        h_dx = h_conv / dx_norm  
  
        self.c_adj = k_dx2 / (k_dx2 + h_dx) # = 0.536  
  
  
  
    def compute_physics_loss(self, xb):  
  
        # BC Loss: T_edge - c_adj*T_adj = 0  
  
        # PDE Loss: Laplacian on interior non-source = 0
```

3.2 Base Model SetFN0Model

Component	Parameter
-----	-----
d_model	256
num_heads	8
n_sab (SetAttention block)	4
fno_ch	64
fno_modes	24
n_fno	6
Total Parameters	**42,830,209 (~42.8M)**

3.3 Training Configuration

Phase 1 (data only, no physics loss):

```
python train_plan_a_physics_v2.py \
    --count-sweep-params ../training_data/params_count_sweep.npy \
    --count-sweep-temps ../training_data/temps_count_sweep.npy \
    --physics-norm --t-ambient 25.0 \
    --d-model 256 --num-heads 8 --n-sab 4 --fno-ch 64 --fno-modes 24 --n-fno 6 \
    --lambda-pde 0.0 --lambda-bc 0.0 \
    --epochs 500 --batch-size 32 --lr 1e-4 \
    --early-stopping --patience 200 \
    --out-dir ./results_plan_a_physics --model-out plan_a_physics_phase1.pth
```

Phase 2 (+ physics constraint):

```
python train_plan_a_physics_v2.py \
    --count-sweep-params ../training_data/params_count_sweep.npy \
    --count-sweep-temps ../training_data/temps_count_sweep.npy \
    --physics-norm --t-ambient 25.0 \
    --d-model 256 --num-heads 8 --n-sab 4 --fno-ch 64 --fno-modes 24 --n-fno 6 \
    --lambda-pde 0.001 --lambda-bc 0.01 \
    --epochs 2000 --batch-size 32 --lr 5e-5 \
    --early-stopping --patience 200 \
    --out-dir ./results_plan_a_physics --model-out plan_a_physics_phase2.pth
```

4. Training Results

4.1 Phase 1 (Data Only)

- Epochs: 500
- Early Stopping: No (data training only)
- Final Train Loss: ~0.04

4.2 Phase 2 (+Physics Loss)

- Early Stopping Epoch: 1510 (patience=200 triggered)
- Best Val Loss: 0.0155
- Loss Breakdown:
- Data Loss: ~0.002
- PDE Loss: ~0.03
- BC Loss: ~0.14

4.3 Training Script Evaluation Bug

The training script has a bug: when computing test R2, it did not inverse-transform predictions, comparing normalized values directly with raw temperatures. This caused summary.json r2_mean = -168.57, True generalization results come from the fixed test scripts predict_plan_a_physics_gen.py and plot_gen_comparison.py (with correct inverse transform applied).

5. Generalization Test Results

Test dataset: temperature_prediction/data/generation_dataset/

Components	Power	Samples	R2 (Plan B)	R2 (Plan A no physics)	Improvement
-----	-----	-----	-----	-----	-----
6 components	15.7W	10	0.9228	0.9068	+0.016
7 components	17.5W	10	0.9101	0.8797	+0.030
8 components	20.0W	10	0.8981	0.8620	+0.036
9 components	30W	10	0.4449	0.0760	+0.369

5.1 Per-Sample Detailed Results (6 components)

Sample	R2	Pred Range	True Range
-----	-----	-----	-----
1	0.9867	[68.9, 82.6]	[69.5, 81.0]
2	0.9366	[69.9, 79.0]	[71.0, 79.0]
3	0.8290	[72.7, 77.4]	[73.0, 77.9]
4	0.9622	[68.9, 82.9]	[70.0, 83.4]
5	0.9052	[70.3, 81.8]	[72.0, 81.7]
6	0.9786	[68.7, 80.0]	[70.4, 79.7]
7	0.9001	[72.3, 80.4]	[72.7, 80.2]
8	0.9776	[69.6, 85.4]	[71.0, 83.7]
9	0.8654	[73.1, 78.8]	[73.5, 77.9]
10	0.8863	[72.2, 78.3]	[72.5, 78.1]

5.2 Per-Sample Detailed Results (9 components)

Sample	R2	Pred Range	True Range
-----	-----	-----	-----
1	0.6177	[112.1, 130.5]	[112.8, 134.5]
2	0.9175	[114.3, 131.1]	[115.0, 132.2]
3	0.0031	[114.1, 129.4]	[115.8, 129.3]
4	0.5966	[112.3, 134.6]	[113.9, 130.7]
5	0.0002	[115.0, 129.5]	[117.5, 130.7]
6	0.2485	[113.2, 129.2]	[114.0, 132.4]
7	0.4060	[115.7, 126.0]	[114.5, 133.3]
8	0.7770	[109.6, 133.3]	[112.1, 132.2]
9	0.1922	[113.5, 128.7]	[115.7, 131.6]
10	0.6897	[114.7, 129.4]	[114.7, 134.4]

6. Key Findings

6.1 Effect of Physics Constraint

- 6/7/8 components: stable improvement of ~2-4%
- 9 components (30W): huge improvement, R2 from 0.076 to 0.445, ~5x improvement
- Physics loss helps model learn physical laws of boundary conditions instead of memorizing training d

6.2 Remaining Issues with 9 Components

- Some samples have extremely low R2 (0.0002, 0.0031)
- 30W power (2-3x outside training range 2.5-13.7W) is still extreme extrapolation
- Temperature range 110-135C with steep gradients near heat sources

6.3 BC Loss vs Data Loss

- Phase2 training: Data Loss ~ 0.002 , BC Loss ~ 0.14
- Physics loss magnitude $\sim 70\times$ larger than data loss
- $\lambda_{bc}=0.01$ makes weighted contributions comparable (BC contribution ~ 0.0014)

7. File Structure

```
temperature_prediction/model_v3/
├─ train_plan_a_physics_v2.py      # Training script (Plan B main)
├─ predict_plan_a_physics_gen.py  # Generalization test script
├─ plot_gen_comparison.py         # GT vs Pred comparison generator
├─ results_plan_a_physics/
|   ├─ plan_a_physics_phase1.pth  # Phase1 model
|   ├─ plan_a_physics_phase2.pth  # Phase2 final model - USE THIS
|   └─ gen_comparison/           # Correct GT/Pred comparison images
|       └─ 6_Component_15.7W/
|           └─ 7_Component_17.5W/
|               └─ 8_Component_20.0W/
|   └─ loss_curves.png
|   └─ r2_scores.png
|   └─ scatter_pred_vs_true.png
|       └─ TECH_REPORT.pdf
└─ models/
    └─ set_fno_thermal.py        # SetFNOModel definition
```

8. Visualization

8.1 Generated During Training (affected by bug, not true performance)

- loss_curves.png - Training/validation loss curves (log10 scale)
- r2_scores.png - Test R2 score distribution
- scatter_pred_vs_true.png - Predicted vs true temperature scatter
- sample_01~06.png - 6 sample heatmap comparisons

8.2 Generalization Test (CORRECT)

- gen_comparison/6_Component_15.7W/sample_01~10.png
- gen_comparison/7_Component_17.5W/sample_01~10.png
- gen_comparison/8_Component_20.0W/sample_01~10.png



9. Known Bug

train_plan_a_physics_v2.py Evaluation Bug

Problem: Test R2 computation during training did not apply inverse transform.

Phenomenon: summary.json r2_mean = -168.57.

Impact: Only affects training script evaluation output. Model weights are unaffected. True generalization results are from predict_plan_a_physics_gen.py and gen_comparison/.

Fix: Apply inverse_transform_temps() before R2 computation (already fixed in predict_plan_a_physics_gen.py and plot_gen_comparison.py).



10. Future Optimization Suggestions

- 1. Fix training script evaluation: add inverse transform before R2
- 2. Try smaller lambda_bc: currently 0.01, try 0.005 or 0.001
- 3. Add in-domain generalization data: include 6-8 component samples in training (not test)
- 4. Expand power range: add 15-20W data to training
- 5. Adjust heat source mask threshold: currently 0.06
- 6. Try Plan A with dropout regularization

11. Conclusion

Physics constraints as auxiliary loss are an effective method to improve power extrapolation generalization. Plan B maintains Plan A's excellent architecture while significantly improving performance

Core Innovation: BC equation precisely derived from SOR solver discrete form, simplified after normalization to $T_{\text{edge}} = 0.536 * T_{\text{adj}}$, with clear physical meaning.